Predictive Analytics for Smart City Infrastructure

M.Tech Project Work Final Review

Presented by

S K Sricharan (25MML1010)

M.Tech Computer Science and Engineering

(Artificial Intelligence and Machine Learning – In collaboration with LTI Mindtree)

Under the Guidance of

Dr R. Srinivasa Perumal

Associate Professor

School of Computer Science and Engineering

Vellore Institute of Technology, Chennai.

CONTENTS

- ABSTRACT
- INTRODUCTION
- LITERATURE SURVEY
- RESEARCH GAPS
- OBJECTIVES
- RESEARCH WORK DONE
  - MODULE IDENTIFICATION
  - SYSTEM ARCHITECTURE
  - DEVELOPMENT & INTEGRATION
  - PROTOTYPE / RESEARCH FRAMEWORK
- CONCLUSIONS
- OUTCOME
- RESULTS & OUTCOMES
- REFERENCES

1.ABSTRACT

Traffic control rooms can see congestion as it happens, but rarely receive forecasts that are explained, honest about their uncertainty and tested before an action is suggested. This project builds a decision-support prototype on public New York City data. A context-gated GATv2–Transformer forecasts link speeds 15, 30 and 60 minutes ahead on a graph of 16 Manhattan DOT speed links (January–October 2026), fusing weather, 311 and calendar context through an availability-aware gate. Split-conformal intervals quantify uncertainty, grouped SHAP explains the trained model, and a LangGraph workflow uses a local LLM (qwen2.5 7B) with FAISS retrieval to draft signal advisories that deterministic safety rules and a SUMO simulation must pass before human review. On a 45-day test period the model reaches 4.77, 5.33 and 6.15 mph MAE, 7–16 % below persistence; intervals reach 90 % coverage overall, and retrieval removed every fabricated citation. An audit of the simulation testbed showed it cannot yet support a controller comparison, which is therefore reported as not run.

2. INTRODUCTION

- Urban congestion depends on demand, neighbouring roads, weather, events and incidents, yet control rooms mostly see only current conditions.
- Graph neural networks forecast traffic on road networks [1]–[3], but give point forecasts without uncertainty or explanation.
- An LLM can phrase advice convincingly without evidence; retrieval grounding [4] and simulation testing [5] are needed before any signal action.
- Objective: join forecasts, uncertainty, explanations, cited evidence and simulation checks in one auditable, human-in-the-loop advisory pipeline on real NYC data.

[1] Y. Li, R. Yu, C. Shahabi, and Y. Liu, “Diffusion convolutional recurrent neural network: Data-driven traffic forecasting,” in Proc. ICLR, 2018.
[2] B. Yu, H. Yin, and Z. Zhu, “Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting,” in Proc. IJCAI, 2018, pp. 3634–3640.
[3] Z. Wu, S. Pan, G. Long, J. Jiang, and C. Zhang, “Graph WaveNet for deep spatial-temporal graph modeling,” in Proc. IJCAI, 2019, pp. 1907–1913.
[4] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in Proc. NeurIPS, 2020.
[5] P. A. Lopez et al., “Microscopic traffic simulation using SUMO,” in Proc. IEEE ITSC, 2018, pp. 2575–2582.

3.LITERATURE SURVEY

S.No	Article Title	Techniques Used	Key Findings	Limitations
1	[1] Li et al., Diffusion Convolutional Recurrent Neural Network (ICLR 2018)	Diffusion convolution on a directed road graph inside a GRU encoder–decoder (seq2seq)	Modelling traffic as diffusion on the directed graph improved multi-step speed forecasts over non-graph baselines	Fixed, predefined graph; point forecasts only; no exogenous context or uncertainty
2	[2] Yu et al., Spatio-Temporal Graph Convolutional Networks (IJCAI 2018)	Graph convolution combined with gated temporal convolution; fully convolutional, no recurrence	Accurate short-term forecasts with faster training and fewer parameters than recurrent models	Static graph; point forecasts; no weather or event context
3	[3] Wu et al., Graph WaveNet (IJCAI 2019)	Self-adaptive adjacency matrix learned from data plus dilated causal convolutions	Captures hidden spatial dependencies even when the given graph is incomplete; long receptive field	Learned graph is hard to interpret; no uncertainty estimates
4	[6] Brody et al., How Attentive are Graph Attention Networks? (ICLR 2022)	GATv2: reorders the attention computation so that attention becomes dynamic	Shows that GAT attention is static; GATv2 is strictly more expressive and performs better on benchmarks	General-purpose graph layer; no temporal modelling

[1] Y. Li, R. Yu, C. Shahabi, and Y. Liu, “Diffusion convolutional recurrent neural network: Data-driven traffic forecasting,” in Proc. ICLR, 2018.
[2] B. Yu, H. Yin, and Z. Zhu, “Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting,” in Proc. IJCAI, 2018, pp. 3634–3640.
[3] Z. Wu, S. Pan, G. Long, J. Jiang, and C. Zhang, “Graph WaveNet for deep spatial-temporal graph modeling,” in Proc. IJCAI, 2019, pp. 1907–1913.
[6] S. Brody, U. Alon, and E. Yahav, “How attentive are graph attention networks?,” in Proc. ICLR, 2022.

3.LITERATURE SURVEY (CONTD.)

S.No	Article Title	Techniques Used	Key Findings	Limitations
5	[7] Angelopoulos & Bates, A Gentle Introduction to Conformal Prediction (2021)	Split conformal prediction wrapped around any trained model	Distribution-free, finite-sample marginal coverage guarantee	Assumes exchangeable data; guarantees marginal, not conditional, coverage
6	[8] Xu & Xie, Conformal Prediction Interval for Dynamic Time-Series (ICML 2021)	EnbPI: bootstrap ensemble with conformal-style intervals for time series	Approximately valid intervals without exchangeability, under stated assumptions on the errors	Requires an ensemble of models; validity is approximate
7	[9] Lundberg & Lee, A Unified Approach to Interpreting Model Predictions (NeurIPS 2017)	SHAP: Shapley-value feature attribution (KernelSHAP and model-specific variants)	Unifies additive attribution methods with consistency guarantees	Cost grows with the number of features; attributions describe the model, not causes
8	[4] Lewis et al., Retrieval-Augmented Generation (NeurIPS 2020)	Dense passage retriever combined with a seq2seq generator	More specific and factual generation on knowledge-intensive tasks than a parametric-only model	Bounded by corpus and retriever quality; can still produce unsupported text

[4] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in Proc. NeurIPS, 2020.
[7] A. N. Angelopoulos and S. Bates, “A gentle introduction to conformal prediction and distribution-free uncertainty quantification,” arXiv:2107.07511, 2021.
[8] C. Xu and Y. Xie, “Conformal prediction interval for dynamic time-series,” in Proc. ICML, PMLR vol. 139, 2021, pp. 11559–11569.
[9] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in Proc. NeurIPS, 2017.

4. RESEARCH GAPS

G1 Forecast stops at the number

Graph forecasting studies [1]–[3] report accuracy but do not connect the forecast to a checked, testable action.

G2 Uncertainty does not decide

Point forecasts dominate; interval width and data quality rarely decide whether an advisory is issued.

G3 Surrogate explanations

Explanations are often computed on simpler surrogate models instead of the deployed forecaster.

G4 Ungrounded LLM advice

LLM recommendations are rarely tied to cited, allow-listed evidence or bounded by deterministic safety rules.

G5 Missing context is hidden

Dead sensors and unavailable feeds are usually imputed silently instead of being modelled with availability masks.

G6 Untested testbeds

Simulation studies seldom audit route choice, demand and signal timing before comparing controllers.

5. OBJECTIVES

O1 Forecast link speeds 15, 30 and 60 min ahead with a context-gated GATv2–Transformer and compare it with six baselines on identical data

Addresses: G1, G5

O2 Quantify uncertainty with per-horizon split-conformal intervals and defer advisories when reliability is low

Addresses: G2

O3 Explain the trained model itself with grouped SHAP and sensitivity tests

Addresses: G3

O4 Draft advisories with a local LLM that must cite retrieved evidence and pass deterministic safety checks

Addresses: G4

O5 Test admissible advisories in a SUMO simulation, after verifying the testbed itself

Addresses: G6

O6 Make every result reproducible and traceable: tests, MLflow, Docker, evidence registry and a paper generated from artifacts

Addresses: all

6. RESEARCH WORK DONE

- The research work is organised in four parts:
  - ❖ **MODULE IDENTIFICATION**
  - ❖ **SYSTEM ARCHITECTURE**
  - ❖ **DEVELOPMENT & INTEGRATION**
  - ❖ **PROTOTYPE / RESEARCH FRAMEWORK**

1,968,303
raw speed records processed

16
Manhattan links modelled

7
forecasting models compared

82
automated tests passing

MODULE IDENTIFICATION

M1 Data ingestion
NYC Open Data, Open-Meteo, OSM; resumable downloads with SHA-256 provenance

M2 Preprocessing & features
5-minute grid, causal lags, availability masks, leakage-tested splits

M3 Road graph
Directed DOT-link graph (26 nodes, 16 edges) as a PyG Temporal signal [10]

M4 Forecasting
Context-gated GATv2–Transformer plus six baselines at 15/30/60 min

M5 Uncertainty
Per-horizon split-conformal intervals and a reliability gate

M6 Explainability
Grouped KernelSHAP on the trained model; neighbour and window sensitivity

M7 Knowledge & RAG
15 approved documents, MiniLM embeddings, FAISS index

M8 LLM agent
LangGraph workflow; local qwen2.5 7B via Ollama; schema-validated JSON

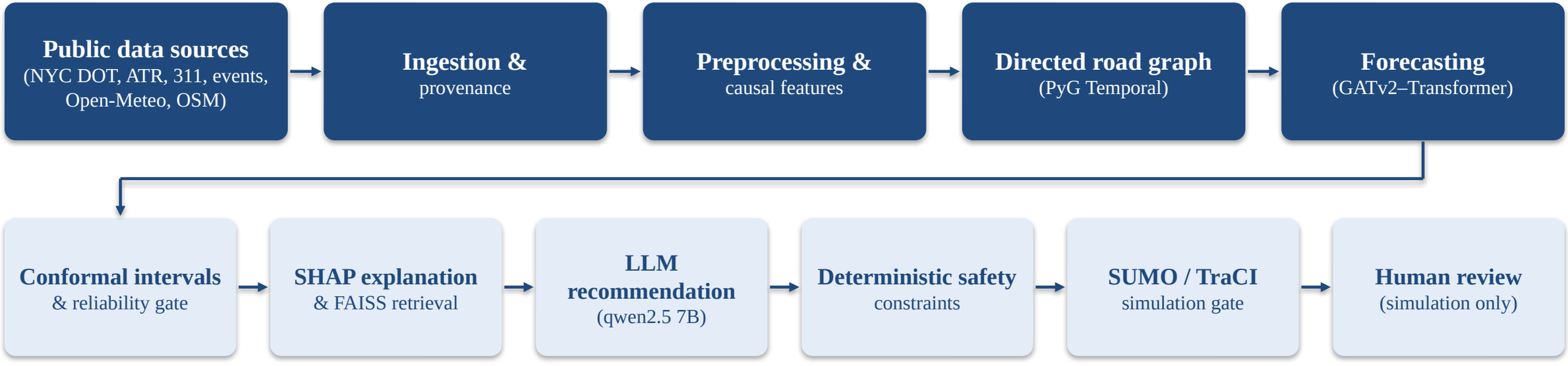
M9 Safety constraints
13 deterministic checks; protected TraCI signal controller

M10 Simulation
SUMO 1.28 West Street corridor; matched seeds; testbed audit

M11 Services
FastAPI, PostgreSQL/PostGIS job queue, worker, Streamlit dashboard

M12 Tracking & reporting
MLflow, evidence registry, claims ledger, paper tables from artifacts

SYSTEM ARCHITECTURE



Data flow

Raw API responses stored verbatim with hashes → 5-minute grid per link → windows of 24 steps × nodes × features → forecasts with 90 % intervals → advisory state in LangGraph → decision and audit trail persisted in PostgreSQL/PostGIS; every run logged in MLflow.

Technology stack

Python 3.12 · Pandas/Polars · PyTorch + PyTorch Geometric + PyG Temporal · XGBoost/LightGBM · SHAP · sentence-transformers + FAISS · Ollama (qwen2.5:7b) · LangGraph · SUMO 1.28 / TraCI · FastAPI · Streamlit · PostgreSQL/PostGIS · MLflow · Docker Compose · Pytest

DEVELOPMENT & INTEGRATION

Implementation of modules

One Python package with a staged command-line interface, YAML configuration and Pydantic contracts

Every stage writes artifacts with run IDs and SHA-256 hashes

The same code path runs the smoke tests and the full experiment

Integration strategy

Shared contracts frozen first: forecast, uncertainty, evidence, candidate action, decision

A stage runner executes the real-data pipeline in fixed order with a stage log

Only the recommendation node calls the LLM; every gate is deterministic code

82 automated tests: unit, leakage, safety, integration, end-to-end (CI run 65, 2026-10-21)

Challenges faced → solutions

7 of 26 links report zero speed all year
→ treated as missing; 16 links kept by a train-coverage rule

DOT link geometries stop short of link ends
→ three documented graph-edge rules

GRU crash with cuDNN on Windows/CUDA
→ cuDNN disabled for GRU runs only

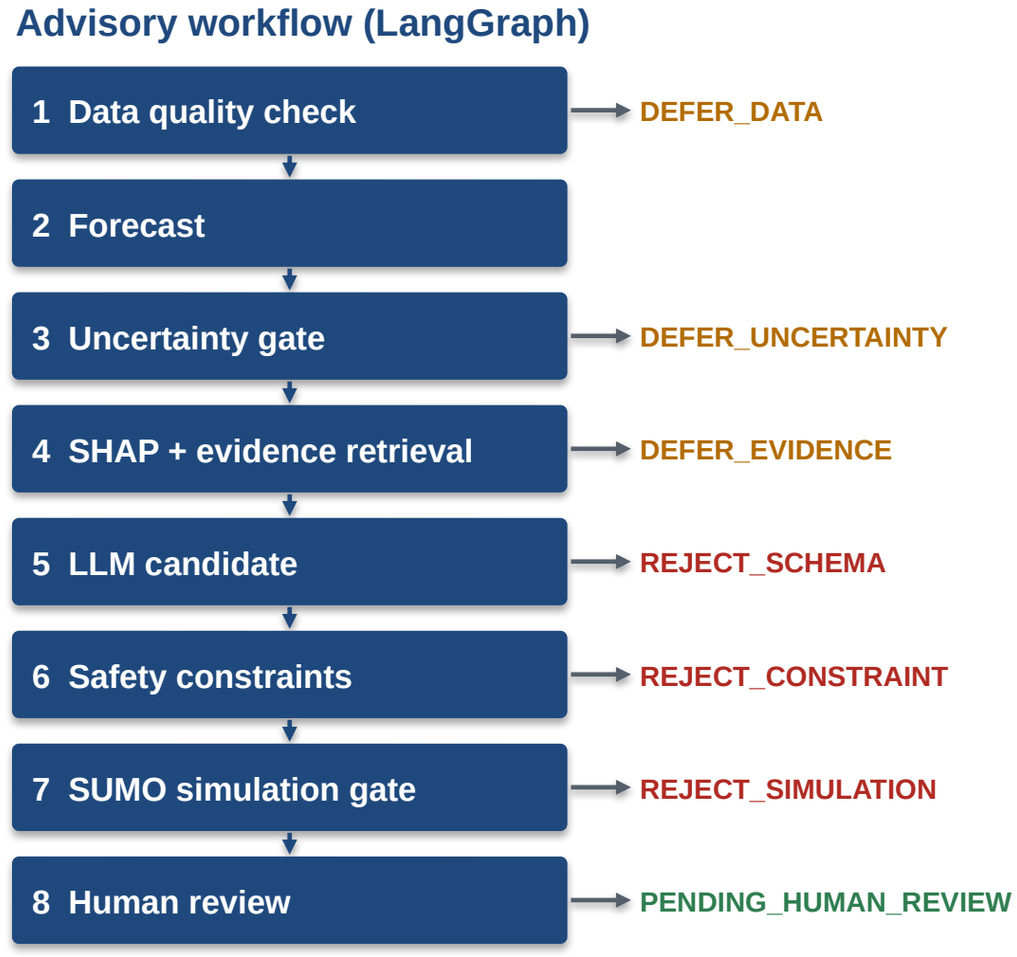
Bootstrap resampled 5-minute steps, not days
→ unit fixed and ablations re-run

A missing field silently blocked every LLM call
→ fixed and agent evaluation re-run

SUMO re-routed half the corridor traffic
→ fixed routes and a testbed audit

PROTOTYPE / RESEARCH FRAMEWORK

- **Command-line pipeline:** one command per stage, from data audit to paper tables
- **FastAPI service:** 18 endpoints for forecasts, explanations, retrieval, advisories, simulations and review; long jobs run in a PostgreSQL-backed worker
- **Streamlit dashboard:** 11 pages, from data audit and traffic map to forecasts, explanations, agent traces, SUMO results and the model card
- **Every view labels its mode:** LIVE, REPLAY or SIMULATION
- **Docker Compose:** PostGIS, API, worker and dashboard images, built and smoke-tested
- **Human review** approves or rejects for simulation only; nothing is sent to real signals



7. CONCLUSIONS

Performance metrics (test period 6 Sep – 20 Oct 2026)

Metric	15 min	30 min	60 min
Proposed model MAE (mph)	**4.77**	**5.33**	**6.15**
Proposed, mean of 3 seeds (mph)	4.69	5.34	6.15
Persistence MAE (mph)	5.14	6.06	7.29
90 % interval coverage	0.909	0.907	0.903
Coverage under congestion	0.798	0.794	0.757
MAE kept after gate (≈26 % deferred)	3.72	4.38	5.25

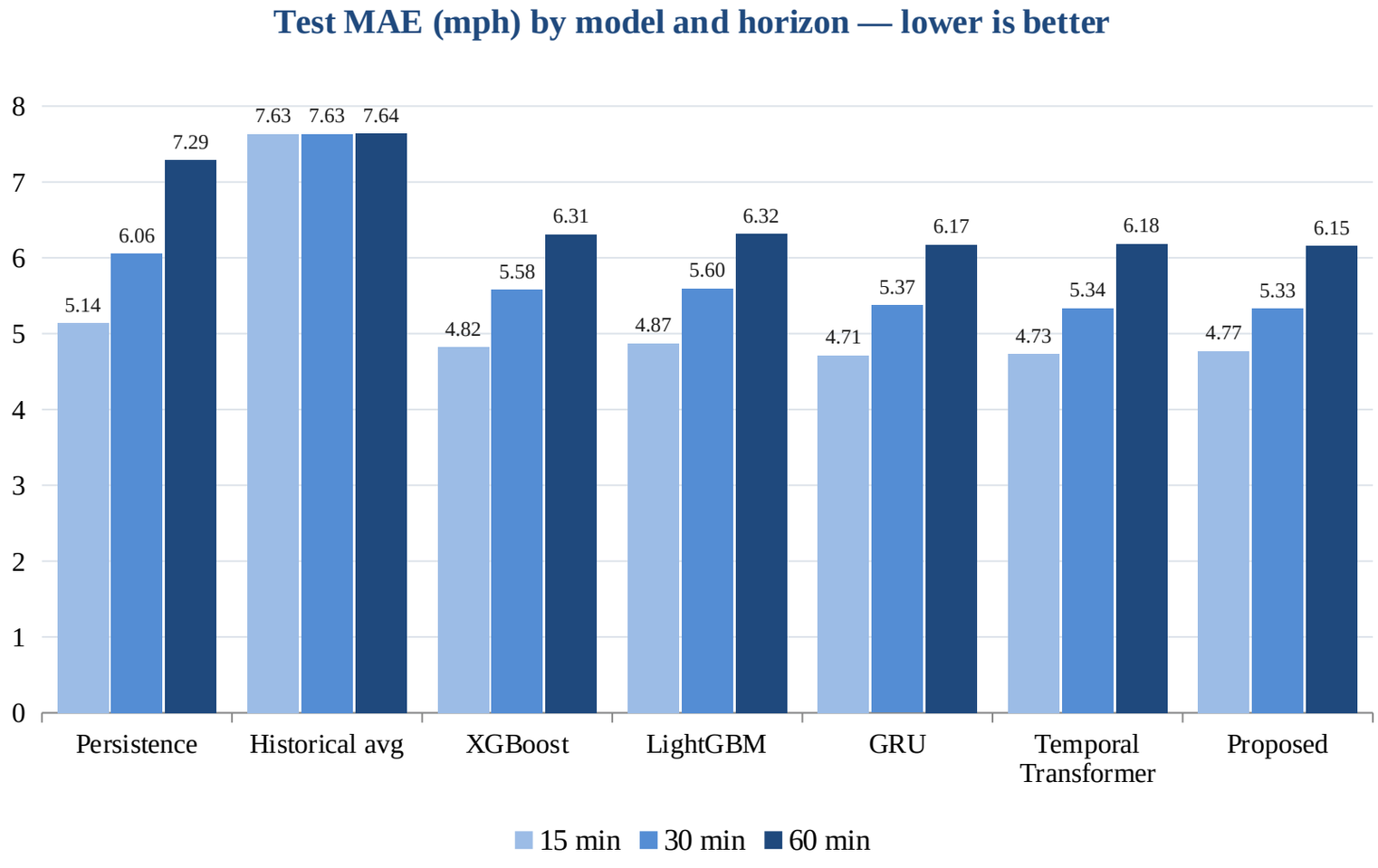
Validation of approach

Chronological splits with purge gaps and leakage tests; identical targets and masks for every model; day-block bootstrap for ablations; a preregistered simulation baseline test; failed runs kept and reported.

Observations

- The proposed model beats persistence by 7–16 % and both tree models. Over three seeds it is 0.03–0.08 mph below the temporal Transformer at every horizon and the GRU at 30/60 min; the GRU is better at 15 min. Small, not decisive.
- The availability-aware gate helps (MAE +0.067 mph when removed); operational weather features hurt (−0.046 mph), a negative result.
- Intervals meet 90 % overall but under-cover congestion, the regime that matters for advisories.
- Retrieval removed fabricated citations (3 of 3 without it, 0 of 9 with it); the constraint layer, not the LLM, decides admissibility.
- The simulation testbed fails its own baseline test, so the controller comparison is not run.

8. OUTCOME



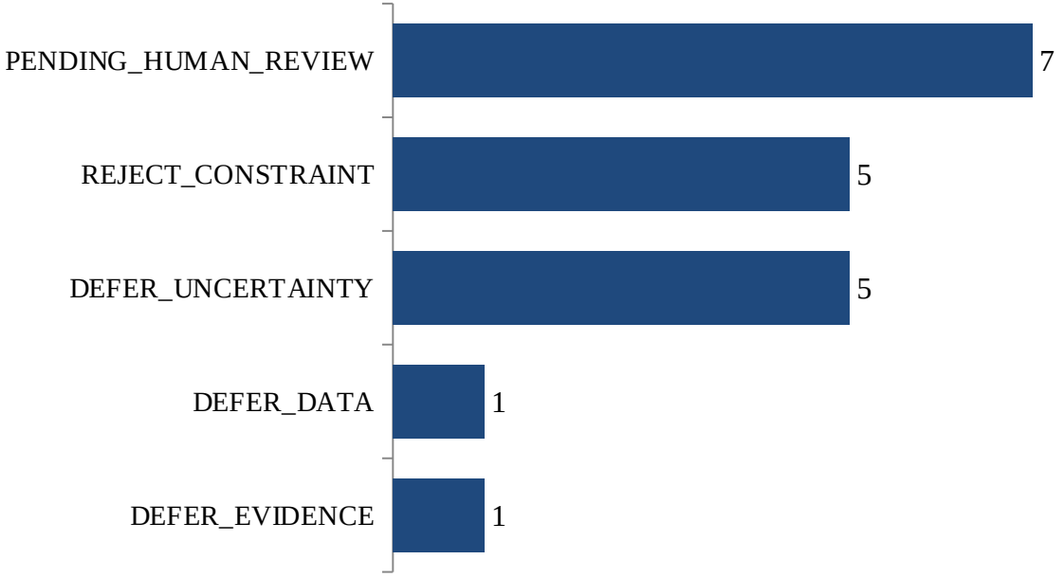
Key results

- Every learned model beats persistence and the historical average.
- Proposed model: 4.77 / 5.33 / 6.15 mph; lowest at 30 and 60 min, GRU lowest at 15 min.
- R^2 rises from 0.55 / 0.46 / 0.30 (persistence) to 0.69 / 0.63 / 0.56.
- 144,205–144,217 test targets per horizon, identical for every model.

Data: NYC DOT speeds [11], 1 Jan – 20 Oct 2026.

8. OUTCOME (CONTD.)

Advisory decisions in 19 workflow runs (REPLAY cases)



12 of 12 LLM responses were schema-valid. Without retrieval the model cited non-existent evidence in 3 of 3 cases (all rejected); with retrieval in 0 of 9.

SUMO testbed outcome (simulation only)

Protocol	Unfinished*	Result
First comparison, 20 Oct (45 runs)	—	**Withdrawn**
2014 count, generated signal timing	17.7 %	**Fails test**
2014 count, Webster re-timed signals	4.3 %	**Fails test**
2025 counts, Webster, no turn-lane limits	10.1 %	**Fails test**

* Vehicles not finishing with no side-street traffic; the preregistered test allows at most 1 % and no stuck vehicles.

The first comparison let SUMO re-route 1,120 of 2,189 main-road vehicles off West Street, so it did not test the corridor.

With real 2025 counts (2,510 NB / 2,444 SB veh/h) the modelled link runs at 5.8 mph; the real street runs at 10.95 mph in the PM peak.

The test was not relaxed: the comparison is reported as not run, and 10.95 mph is now the calibration target.

8. OUTCOME: WORKING DASHBOARD (1/4)

Overview & status (status page, no mode label)

Multi-horizon traffic forecasting & explainable agentic signal advisory

Predictive Analytics for Smart City Infrastructure - Multi-Horizon Traffic Congestion Forecasting and Explainable Agentic Signal Advisory Using Spatiotemporal Transformers and RAG

Proposed model	Track	Study area
proposed-20261020T…	operational	manhattan_dot_spee…

Component status (evidence registry)

requirement_id	component	test_status	artifact_path	limitations
DATA-01	data	VERIFIED	reports/data_audit/source_inventory.csv	None
DATA-02	data	PARTIAL	data/manifests/source_manifest.json	The forecasting data (sp
DATA-03	data	VERIFIED	reports/data_audit/availability_audit.md	None
GRAPH-01	graph	VERIFIED	reports/data_audit/graph_validation.json	None
GRAPH-02	graph	VERIFIED	artifacts/models/proposed-20261020T210211-4a0685/feature_schema.json	None
MODEL-01	models	VERIFIED	artifacts/models/registry_operational.json	None
MODEL-02	models	VERIFIED	reports/experiments/forecast_metrics.csv	None
UNC-01	uncertainty	VERIFIED	artifacts/calibration/calibrator.json	None
XAI-01	explainability	VERIFIED	artifacts/explanations/explanations_summary.json	None
RAG-01	rag	VERIFIED	artifacts/retrieval/retrieval_eval.json	None

› final_status.md

Data audit & provenance Mode shown: REPLAY

Data audit & provenance

Mode: REPLAY

source_id	official_title	publisher	dataset_id	endpoint
speeds	DOT Traffic Speeds	City of New York	i4gi-tjb9	https://data.cityofnewyork.us/resource/i4gi-tjb9.json
volume	Automated Traffic Volume Counts	City of New York	7ym2-wayt	https://data.cityofnewyork.us/resource/7ym2-wayt.json
events	NYC Permitted Event Information	Office of Citywide Event Coord	tvpp-9vvx	https://data.cityofnewyork.us/resource/tvpp-9vvx.json
events_hist	NYC Permitted Event Information - Historical	City of New York	bkfu-528j	https://data.cityofnewyork.us/resource/bkfu-528j.json
collisions	Motor Vehicle Collisions - Crashes	Police Department (NYPD)	h9gi-nx95	https://data.cityofnewyork.us/resource/h9gi-nx95.json
sr311	311 Service Requests from 2020 to Present	311	erm2-nwe9	https://data.cityofnewyork.us/resource/erm2-nwe9.json
weather	Open-Meteo archive (ERA5-based reanalysis) + historical forecast API	Open-Meteo		https://archive-api.open-meteo.com/v1/archive;https://historic
osm	OpenStreetMap extracts via Overpass API	OpenStreetMap contributors		https://overpass-api.de/api/interpreter

Coverage by year (server-side counts)

source_id	dataset_id	year	record_count
speeds	i4gi-tjb9	1930	8
speeds	i4gi-tjb9	1978	15
speeds	i4gi-tjb9	2014	6
speeds	i4gi-tjb9	2015	9
speeds	i4gi-tjb9	2016	36
speeds	i4gi-tjb9	2017	6019989
speeds	i4gi-tjb9	2018	13860107
speeds	i4gi-tjb9	2019	12971855
speeds	i4gi-tjb9	2020	11943235
speeds	i4gi-tjb9	2021	12203801

Screenshots of the project's Streamlit dashboard (src/smartcity/dashboard/app.py) captured on 21 Oct 2026; every number on them is read from committed result files. Live NYC feeds were not reachable from the capture machine, so no LIVE view is shown.

8. OUTCOME: WORKING DASHBOARD (2/4)

Experiment results Mode shown: REPLAY

Experiment results

Mode: REPLAY

forecast_metrics

run_id	model	track	split	evaluation_period_utc	horizon_min	n_valid_targets	mae_mp
persistence-20261020T193923-c48849	persistence	operational	test	2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00	15	144217	5.13860
persistence-20261020T193923-c48849	persistence	operational	test	2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00	30	144213	6.05562
persistence-20261020T193923-c48849	persistence	operational	test	2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00	60	144205	7.28756
historical_average-20261020T193932-e08975	historical_average	operational	test	2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00	15	144217	7.63218
historical_average-20261020T193932-e08975	historical_average	operational	test	2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00	30	144213	7.63295
historical_average-20261020T193932-e08975	historical_average	operational	test	2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00	60	144205	7.63695
xgboost-20261020T193933-18d124	xgboost	operational	test	2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00	15	144217	4.82294
xgboost-20261020T193933-18d124	xgboost	operational	test	2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00	30	144213	5.57922
xgboost-20261020T193933-18d124	xgboost	operational	test	2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00	60	144205	6.30799
lightgbm-20261020T193956-ccbf73	lightgbm	operational	test	2026-09-06 09:00:00+00:00 .. 2026-10-20 04:00:00+00:00	15	144217	4.87253

ablation_metrics

variant	track	run_id	mae_mean_mph	delta_vs_proposed_mph	ci95_low	ci95_high	n_days	status
proposed	operational	proposed-20261020T201110-0e333f	5.4194	0.0	0.0	0.0	45	REFERENC
ablation_no_gate	operational	ablation_no_gate-20261020T202053-48c9ff	5.4867	0.0673	0.0482	0.0864	44	OK
ablation_no_weather	operational	ablation_no_weather-20261020T202610-295c01	5.3732	-0.0462	-0.0692	-0.0205	44	OK
ablation_no_infrastructure	operational	ablation_no_infrastructure-20261020T203545-6cd844	5.433	0.0136	0.0019	0.0258	44	OK
ablation_no_events	operational	ablation_no_events-20261020T204114-372604						NOT_APPL

Uncertainty & reliability Mode shown: REPLAY

coverage_by_horizon.csv

horizon_min	group	n	coverage	mean_width_mph
15	all	144217	0.909	24.2562
30	all	144213	0.9074	26.945
60	all	144205	0.9033	29.5394

coverage_by_scenario.csv

horizon_min	group	n	coverage	mean_width_mph
15	free_ratio>=0.8	98556	0.9311	24.2562
15	moderate_0.5-0.8	22396	0.9266	24.2562
15	congested<0.5	23265	0.7982	24.2562
30	free_ratio>=0.8	98546	0.9301	26.945
30	moderate_0.5-0.8	22398	0.9259	26.945
30	congested<0.5	23269	0.7935	26.945
60	free_ratio>=0.8	98522	0.9329	29.5394
60	moderate_0.5-0.8	22406	0.9253	29.5394
60	congested<0.5	23277	0.7569	29.5394
15	non_rush	106048	0.8984	24.2562

deferral_tradeoff.csv

missing_threshold	horizon_min	n_valid	retained	deferral_rate	mae_retained	mae_all	coverage_retained
0	15	144217	102928	0.2863	3.5605	4.7727	0.9531

Screenshots of the project's Streamlit dashboard (src/smartcity/dashboard/app.py) captured on 21 Oct 2026; every number on them is read from committed result files. Live NYC feeds were not reachable from the capture machine, so no LIVE view is shown.

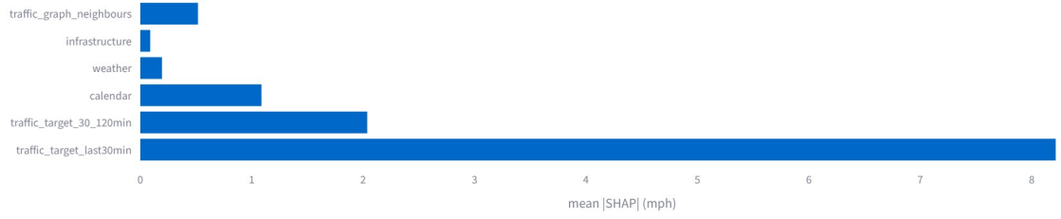
8. OUTCOME: WORKING DASHBOARD (3/4)

Explainability (proposed model) Mode shown: REPLAY

Explanations of the proposed model

Mode: REPLAY

proposed (context-gated GATv2-Transformer) — grouped KernelSHAP; sensitivities, not causal effects.



Local alert

4620298 h=30


```
{
  "explanation_id" : "xai-20261020T212531-006725"
  "method" : "grouped KernelSHAP (exact, all coalitions) on the proposed model"
  "link_id" : "4620298"
  "link_index" : 15
  "horizon_minutes" : 30
  "origin_index" : 83417
  "origin_utc" : 1792269000
  "background" : "16 windows sampled from TRAIN partition (seed 42)"
  "players" : [
```


Agent trace & advisory Mode shown: REPLAY

Agent traces & advisories

run_id	case	decision	reasons
adv-20261021T043410-63bde3	mapped_corridor_rush_hour	PENDING_HUMAN_REVIEW	monitor_only advisory (no signal change proposed)
adv-20261021T043340-e5677a	rain_period	DEFER_UNCERTAINTY	WIDE_INTERVAL: h=15 width 44.4 > 38.1 mph; WIDE_INTERVAL: h=30 width 45.8 > 38.1 m
adv-20261021T043340-9374e9	mapped_corridor_alert	PENDING_HUMAN_REVIEW	monitor_only advisory (no signal change proposed)
adv-20261021T043316-927b0c	routine_rush_hour	REJECT_CONSTRAINT	network_mapping_valid: no reviewed DOT->SUMO mapping
adv-20261021T043252-38b42e	mapped_corridor_rush_hour	PENDING_HUMAN_REVIEW	monitor_only advisory (no signal change proposed)
adv-20261021T043216-ea1e44	rain_period	DEFER_UNCERTAINTY	WIDE_INTERVAL: h=15 width 44.4 > 38.1 mph; WIDE_INTERVAL: h=30 width 45.8 > 38.1 m
adv-20261021T043216-99095f	mapped_corridor_alert	PENDING_HUMAN_REVIEW	monitor_only advisory (no signal change proposed)
adv-20261021T043143-66b3d2	routine_rush_hour	PENDING_HUMAN_REVIEW	monitor_only advisory (no signal change proposed)
adv-20261021T043115-bf3109	mapped_corridor_rush_hour	REJECT_CONSTRAINT	retrieval_support_acceptable: evidence ids not in retrieved set: ['EVIDENCE_123456']
adv-20261021T043049-b4865d	mapped_corridor_alert	REJECT_CONSTRAINT	retrieval_support_acceptable: evidence ids not in retrieved set: ['EVIDENCE_123456']

Trace

adv-20261021T043410-63bde3

Mode: REPLAY

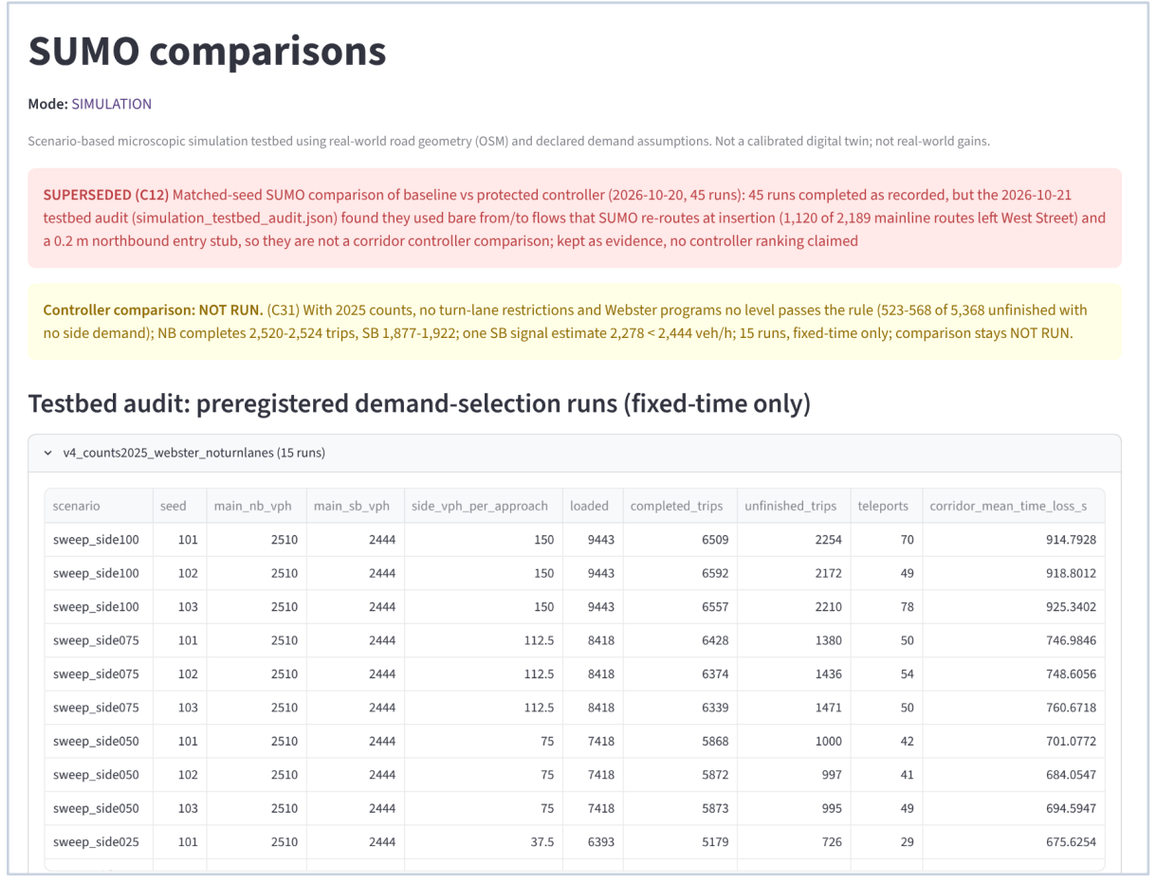
Decision: `PENDING_HUMAN_REVIEW`

- data_quality
- forecast
- uncertainty
- retrieval

Screenshots of the project's Streamlit dashboard (src/smartcity/dashboard/app.py) captured on 21 Oct 2026; every number on them is read from committed result files. Live NYC feeds were not reachable from the capture machine, so no LIVE view is shown.

8. OUTCOME: WORKING DASHBOARD (4/4)

SUMO comparisons Mode shown: **SIMULATION**



SUMO comparisons

Mode: SIMULATION

Scenario-based microscopic simulation testbed using real-world road geometry (OSM) and declared demand assumptions. Not a calibrated digital twin; not real-world gains.

SUPERSEDED (C12) Matched-seed SUMO comparison of baseline vs protected controller (2026-10-20, 45 runs): 45 runs completed as recorded, but the 2026-10-21 testbed audit (simulation_testbed_audit.json) found they used bare from/to flows that SUMO re-routes at insertion (1,120 of 2,189 mainline routes left West Street) and a 0.2 m northbound entry stub, so they are not a corridor controller comparison; kept as evidence, no controller ranking claimed

Controller comparison: NOT RUN. (C31) With 2025 counts, no turn-lane restrictions and Webster programs no level passes the rule (523-568 of 5,368 unfinished with no side demand); NB completes 2,520-2,524 trips, SB 1,877-1,922; one SB signal estimate 2,278 < 2,444 veh/h; 15 runs, fixed-time only; comparison stays NOT RUN.

Testbed audit: preregistered demand-selection runs (fixed-time only)

v4_counts2025_webster_noturnlanes (15 runs)

scenario	seed	main_nb_vph	main_sb_vph	side_vph_per_approach	loaded	completed_trips	unfinished_trips	teleports	corridor_mean_time_loss_s
sweep_side100	101	2510	2444	150	9443	6509	2254	70	914.7928
sweep_side100	102	2510	2444	150	9443	6592	2172	49	918.8012
sweep_side100	103	2510	2444	150	9443	6557	2210	78	925.3402
sweep_side075	101	2510	2444	112.5	8418	6428	1380	50	746.9846
sweep_side075	102	2510	2444	112.5	8418	6374	1436	54	748.6056
sweep_side075	103	2510	2444	112.5	8418	6339	1471	50	760.6718
sweep_side050	101	2510	2444	75	7418	5868	1000	42	701.0772
sweep_side050	102	2510	2444	75	7418	5872	997	41	684.0547
sweep_side050	103	2510	2444	75	7418	5873	995	49	694.5947
sweep_side025	101	2510	2444	37.5	6393	5179	726	29	675.6254

What the dashboard shows, and what it does not

- The SUMO page states the first 45-run comparison is SUPERSEDED and the controller comparison NOT RUN, so it shows no simulated gain.
- Agent traces end in PENDING_HUMAN_REVIEW, DEFER or REJECT; approval is for simulation only and nothing is sent to real signals.
- Not captured yet: the Multi-horizon forecast and Traffic map pages need the trained model file and processed data held on the training laptop; they will be captured there for the final version.
- The Experiment results page shows the seed-42 runs; the three-seed results are on slide 21.

Screenshots of the project's Streamlit dashboard (src/smartcity/dashboard/app.py) captured on 21 Oct 2026; every number on them is read from committed result files. Live NYC feeds were not reachable from the capture machine, so no LIVE view is shown.

9. RESULTS & OUTCOMES

Held-out test period 6 Sep – 20 Oct 2026, 16 Manhattan links; advisory and retrieval results from the REPLAY workflow runs

Test MAE (mph): mean ± s.d. over seeds 42, 1 and 2

Model	15 min	30 min	60 min
Proposed GATv2–Transformer	**4.69 ± 0.06**	**5.34 ± 0.03**	**6.15 ± 0.02**
GRU	4.67 ± 0.06	5.37 ± 0.02	6.18 ± 0.01
Temporal Transformer	4.72 ± 0.03	5.38 ± 0.03	6.23 ± 0.08
Proposed, seed 42 only	4.77	5.33	6.15
Persistence	5.14	6.06	7.29

Paired day-block bootstrap (44 test days): the proposed model is lower than the temporal Transformer at every horizon (0.035 / 0.037 / 0.080 mph) and than the GRU at 30 and 60 min (0.032 / 0.031); the GRU is lower at 15 min (0.021). All intervals exclude zero, but with three seeds the advantage is small, not decisive.

Source: reports/experiments/multi_seed_summary.csv (nine GPU runs, 21 Oct 2026).

7–16 %
lower MAE than persistence (seed 42 run)

0.91 / 0.91 / 0.90
coverage of the 90 % conformal intervals; 0.76–0.80 under congestion

15–22 %
lower MAE on forecasts the reliability gate keeps (≈26 % deferred)

24 / 24
local-LLM responses schema-valid over two workflow passes (qwen2.5 7B)

3 of 3 → 0 of 9
advisories citing non-existent evidence, without vs with retrieval

19
workflow runs per pass: 7 pending human review, 5 rejected by safety checks, 7 deferred

Also measured: removing the availability-aware gate raises MAE by 0.067 mph (95 % CI 0.048–0.086); removing operational weather features lowers it by 0.046 mph (a negative result). No simulated controller gain is claimed: the SUMO testbed fails its own baseline test (5.8 vs 10.95 mph).

9. RESULTS & OUTCOMES (CONTD.)

Implemented and verified

- Real-data pipeline on NYC DOT speeds, 1 Jan – 20 Oct 2026, with provenance hashes
- Seven forecasting models on identical targets; three-seed study of the deep models; ablations
- Conformal intervals, reliability gate and grouped SHAP on the trained model
- FAISS retrieval over 15 approved documents; local LLM with schema validation
- LangGraph workflow with 13 deterministic safety checks
- FastAPI, PostGIS jobs, Streamlit, MLflow, Docker Compose

Partial or not run

- SUMO controller comparison NOT RUN; the first 45-run comparison is superseded
- No real advisory has reached the simulation gate (the LLM chose monitor-only in both passes)
- Only three seeds; the proposed model is not bit-reproducible on GPU at a fixed seed
- Road graph is sparse: 26 nodes, 16 edges
- Intervals under-cover congested periods

Outcomes and next steps

- IEEE conference paper (LaTeX and Word), every claim traced in the claims ledger
- GitHub repository with tests, runbooks and result files
- Next: calibrate West St signal timing to 10.95 mph or obtain NYC DOT timing plans, then run the comparison
- Next: more seeds without the compute budget
- Next: paper submission

REFERENCES

[1] Y. Li, R. Yu, C. Shahabi, and Y. Liu, “Diffusion convolutional recurrent neural network: Data-driven traffic forecasting,” in Proc. ICLR, 2018.

[2] B. Yu, H. Yin, and Z. Zhu, “Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting,” in Proc. IJCAI, 2018, pp. 3634–3640.

[3] Z. Wu, S. Pan, G. Long, J. Jiang, and C. Zhang, “Graph WaveNet for deep spatial-temporal graph modeling,” in Proc. IJCAI, 2019, pp. 1907–1913.

[4] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in Proc. NeurIPS, 2020.

[5] P. A. Lopez et al., “Microscopic traffic simulation using SUMO,” in Proc. IEEE ITSC, 2018, pp. 2575–2582.

[6] S. Brody, U. Alon, and E. Yahav, “How attentive are graph attention networks?,” in Proc. ICLR, 2022.

[7] A. N. Angelopoulos and S. Bates, “A gentle introduction to conformal prediction and distribution-free uncertainty quantification,” arXiv:2107.07511, 2021.

[8] C. Xu and Y. Xie, “Conformal prediction interval for dynamic time-series,” in Proc. ICML, PMLR vol. 139, 2021, pp. 11559–11569.

[9] S. M. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in Proc. NeurIPS, 2017.

[10] B. Rozemberczki et al., “PyTorch Geometric Temporal: Spatiotemporal signal processing with neural machine learning models,” in Proc. ACM CIKM, 2021, pp. 4564–4573.

[11] NYC Department of Transportation, “DOT Traffic Speeds NBE,” NYC Open Data, dataset i4gi-tjb9, accessed Oct. 2026.

[12] NYC Department of Transportation, “Automated Traffic Volume Counts,” NYC Open Data, dataset 7ym2-wayt, accessed Oct. 2026.

Thank You